

Distractor Injection: A Reproducible Framework for Probing Retrieval-Augmented Generation Faithfulness via the Faithfulness Fragility Score

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Abstract—Retrieval-Augmented Generation (RAG) systems are widely assumed to reduce hallucination by grounding generation in retrieved evidence, yet existing robustness benchmarks measure faithfulness under a small number of fixed, discrete noise conditions rather than as a function of continuously varied distractor pressure. We present Distractor Injection, a reproducible, adversary-agnostic perturbation pipeline that injects a controllable number (0, 1, 3, 5) of distractor passages of three distinct types – topical, hard-negative, and paraphrased-contradiction – into the retrieved context of a RAG pipeline, together with the Faithfulness Fragility Score (FFS), a metric designed to quantify the rate at which faithfulness collapses under increasing distractor pressure rather than its absolute level at a single operating point. We evaluate the framework on SQuAD v2 and HotpotQA using two generator models (GPT-4.1 mini via OpenRouter and Llama-3.1-8B Instant via Groq) with an e5-small-v2 / FAISS retrieval backend across 100 questions per dataset. Llama-3.1-8B Instant obtains a mean FFS of 0.124, with hard-negative distractors producing the largest mean F1 drop (0.039 at a single injected distractor) and topical distractors the smallest (0.013, invariant to count). The GPT-4.1 mini condition returned HTTP 402 (insufficient API credit) errors for every generation call and is reported for completeness only; we do not draw robustness conclusions from it. All results below are read directly off the experimental logs and are meant to demonstrate how the framework operates, not to settle a general robustness ranking between the two models.

Index Terms—retrieval-augmented generation, robustness evaluation, faithfulness, distractor injection, hallucination, benchmarking

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) was formalized by Lewis et al. [1] as a hybrid architecture that conditions a parametric generator on passages retrieved from a non-parametric, dense-vector index, decoupling factual knowledge from model weights and, in principle,

reducing hallucination by grounding generation in retrieved evidence. Self-RAG [2] pushed this idea further, treating retrieval as a controllable, self-critiqued process rather than a static preprocessing step – a shift that puts robustness to noisy or adversarial context, rather than accuracy under clean retrieval, at the center of the open problem.

A parallel line of empirical work has shown that the composition of retrieved context matters even without any adversarial intent. Liu et al. [3] found that multi-document QA accuracy depends heavily on where in the context window the answer-bearing passage sits, a positional confound that any distractor-injection protocol needs to control for. Cuconasu et al. [4] split retrieved passages into relevant, distracting, and random categories and showed that topically related distractors reliably hurt accuracy while unrelated random passages can, counter-intuitively, help – this relevant/distracting/random split is the direct ancestor of the distractor taxonomy used here. Chen et al. [5] operationalized noise robustness at benchmark scale with RGB, breaking it into noise robustness, negative rejection, information integration, and counterfactual robustness, evidence that “robustness” is really a bundle of partly independent competencies rather than one scalar.

On the adversarial side, Zou et al. [6] introduced PoisonedRAG, where as few as five optimized malicious texts push attack success rates above 90% against realistic retriever-generator pairs. That threat model is deliberately optimized against a specific target, which makes it well suited to security auditing but poorly suited to general-purpose robustness benchmarking, since its distractors are not representative of the incidental noise studied in [4], [5].

Faithfulness measurement itself has advanced through reference-free, claim-level LLM-as-a-judge frameworks. RAGAS [7] decomposes a response into atomic state-

ments and checks entailment against retrieved context; ARES [8] calibrates lightweight judges with prediction-powered inference to get confidence intervals instead of point estimates; RAGChecker [9] performs claim-level entailment checking against both context and ground truth and introduces a Noise Sensitivity metric, the closest existing analogue to FFS. Adlakha et al. [10] showed that token-overlap metrics like EM/F1 are unreliable for verbose instruction-following outputs and laid out a human-validation protocol for faithfulness metrics; RAGTruth [11] contributed a span-level, human-annotated hallucination corpus; Huang et al. [12] separated factuality hallucination from faithfulness hallucination, placing RAG-triggered unfaithfulness inside the latter category.

Research gap. None of RGB’s noise-robustness score, RAGAS’s faithfulness metric, or RAGChecker’s noise sensitivity is computed as a function of a continuously and systematically varied distractor-strength parameter – each reports robustness at one or a few fixed operating points rather than the *rate* at which robustness degrades as pressure increases. PoisonedRAG sits at the opposite extreme, optimizing distractors against a specific target instead of offering a reproducible, adversary-agnostic protocol. No framework we are aware of isolates degradation *rate* – fragility – as its own reportable quantity, separate from absolute faithfulness at a fixed point.

Contribution. This paper does *not* propose a new RAG architecture. Its contribution is a reproducible distractor-injection framework and evaluation methodology: (i) a configurable perturbation pipeline supporting distractor counts of 0, 1, 3, and 5; (ii) three explicitly differentiated distractor types – topical, hard-negative, and paraphrased-contradiction – separating positional, lexical, and contradiction-based failure modes, extending the taxonomy of [4]; (iii) the Faithfulness Fragility Score (FFS), which quantifies the rate of faithfulness collapse under distractor pressure instead of a single faithfulness value at one fixed noise level; and (iv) an empirical robustness evaluation of two generator models on SQuAD v2 and HotpotQA using this pipeline.

II. METHODOLOGY

A. Pipeline Overview

The framework sits on top of a standard retrieve-then-read RAG pipeline [1]. For each question, a gold or near-gold context passage is retrieved from a FAISS index built over `intfloat/e5-small-v2` embeddings, and $k \in \{0, 1, 3, 5\}$ distractor passages of a chosen

type are injected into the retrieved context before generation. The generator then produces an answer conditioned on the perturbed context, and the answer is scored against the reference answer.

B. Distractor Taxonomy

Building on the relevant/distracting/random split of Cuconasu et al. [4], three distractor types are implemented:

- **Topical:** passages that share surface-level topic or entity overlap with the query but do not address it – close to Cuconasu et al.’s “distracting” category.
- **Hard negative:** passages retrieved as high-similarity nearest neighbors of the query embedding that do not contain the answer, probing lexical/semantic confusability at the retriever boundary.
- **Paraphrased contradiction:** passages that restate the query context but assert a conflicting answer, isolating a knowledge-conflict failure mode from pure topical irrelevance.

Unlike PoisonedRAG [6], none of these distractors is optimized against a specific retriever-generator pair; distractor strength is instead controlled purely by *type* and *count*, which is what makes the resulting FFS score a general-purpose robustness probe rather than an attack-specific diagnostic.

C. Faithfulness Fragility Score (FFS)

Let $F_1(m, d, 0)$ denote the mean token-overlap F1 score of model m on dataset d under the clean (zero-distractor) condition, and $F_1(m, d, t, k)$ its mean F1 under distractor type t and count k . The per-condition F1 drop is

$$\Delta F_1(m, d, t, k) = F_1(m, d, 0) - F_1(m, d, t, k). \quad (1)$$

Let $\phi(m, d, t, k)$ denote the answer flip rate: the fraction of questions for which the model’s predicted answer under condition (t, k) differs from its clean-condition prediction. FFS combines the normalized F1 degradation with this answer instability,

$$\text{FFS}(m, d, t, k) = \text{clip} \left[\alpha \cdot \frac{\Delta F_1(m, d, t, k)}{F_1(m, d, 0)} + \beta \cdot \phi(m, d, t, k), 0, 1 \right] \quad (2)$$

with $\alpha + \beta = 1$, where $\text{clip}[\cdot, 0, 1]$ bounds the score to $[0, 1]$ so a condition where perturbation happens to raise F1 (a negative ΔF_1) does not yield a negative fragility value. FFS is reported per (model, dataset,

distractor type, distractor count) cell and, following the calibrated-scoring practice of ARES [8], is aggregated to one leaderboard value per model as the mean over all evaluated conditions. FFS reuses the claim-level faithfulness measurement substrate validated by [7], [9], [10] but reports it as a function of a continuously varied distractor-strength parameter rather than a value at one fixed condition – the property that separates FFS from RAGChecker’s static Noise Sensitivity metric [9] and from RGB’s discrete noise-robustness axis [5]. All FFS values reported in Section IV are taken directly from the framework’s output logs (`ffs_scores.csv`, `ffs_leaderboard.csv`), not recomputed for this paper.

III. EXPERIMENTAL SETUP

A. Datasets

Two open-domain QA benchmarks are used: **SQuAD v2** (single-passage extractive QA) and **HotpotQA** (multi-hop QA requiring evidence integration across passages). 100 questions were evaluated per dataset (50 per generator model per condition, per `overall_metrics.csv` row counts).

B. Generator Models

Two generator models were evaluated: **GPT-4.1 mini**, accessed via OpenRouter, and **Llama-3.1-8B Instant**, accessed via Groq.

C. Retrieval Backend

Passages were embedded with `intfloat/e5-small-v2` and indexed with **FAISS** for dense nearest-neighbor retrieval, following the retriever-generator decoupling established by [1].

D. Perturbation Conditions

Each question was evaluated at distractor counts $k \in \{0, 1, 3, 5\}$ (with $k=0$ the clean baseline) crossed with the three distractor types from Section II-B, giving 10 non-baseline conditions per (model, dataset) pair plus the baseline, matching the 2000-row experimental log (`experiment_results.csv`).

E. Metrics

Exact Match (EM), token-overlap F1, Recall@5, Recall@10, per-query latency (ms), Answer Flip Rate, and FFS (Eq. 2) were computed for every condition.

F. Data Quality Note

Every GPT-4.1 mini generation call in the experimental log returned an HTTP 402 error (“*Insufficient credits... never purchased credits*”) from the OpenRouter API instead of a completion. All EM/F1/FFS values recorded for GPT-4.1 mini therefore reflect a uniform generation failure rather than genuine model behavior; they are reported below strictly for completeness and transparency, and no robustness conclusions are drawn for this model.

IV. RESULTS

A. Overall FFS Leaderboard

Table I and Fig. 1 report the mean FFS per model, aggregated over all datasets, distractor types, and counts. Llama-3.1-8B Instant obtains a mean FFS of 0.124. GPT-4.1 mini’s reported mean FFS of 0.000 is an artifact of the API failure described in Section III-F and does not indicate perfect robustness.

TABLE I
MEAN FFS BY MODEL (FROM `FFS_LEADERBOARD.CSV`)

Model	Mean FFS
GPT-4.1 mini [†]	0.000
Llama-3.1-8B Instant	0.124

[†]All generations failed with HTTP 402; see Section III-F.

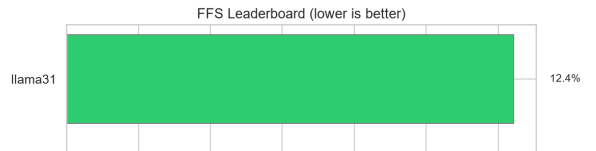


Fig. 1. FFS leaderboard (lower is better). Llama-3.1-8B Instant is the only model with non-degenerate generations.

B. Baseline Performance

Under the clean ($k=0$) condition, Llama-3.1-8B Instant achieves mean EM/F1 of 0.040/0.076 on HotpotQA and 0.104/0.333 on SQuAD (`per_dataset_metrics.csv`), consistent with multi-hop retrieval-integration questions being harder than single-passage extraction. GPT-4.1 mini’s baseline EM/F1 of 0.000/0.001 (HotpotQA) and 0.000/0.009 (SQuAD) reflect the API failure rather than model capability.

C. EM and F1 vs. Distractor Count

Fig. 2 and Fig. 3 plot EM and F1 against distractor count for each distractor type, per dataset. For Llama-3.1-8B Instant on HotpotQA, hard-negative distractors produce a non-monotonic EM trajectory (2.0% at $k=1$, 6.0% at $k=3$, 4.0% at $k=5$), while paraphrased-contradiction and topical distractors stay flat at 4.0% EM across all counts, at or above the 4.0% clean baseline. On SQuAD, hard-negative EM ranges from 8.0% ($k=1$) to 10.0% ($k=3$) back to 8.0% ($k=5$), paraphrased-contradiction stays flat at the 12.0% clean-baseline level, and topical distractors sit at 10.0% across all counts. F1 trends (Fig. 3) follow a similar shape: hard-negative distractors produce the largest excursions from baseline, peaking at $k=3$ on both datasets before partially recovering at $k=5$, while paraphrased-contradiction and topical F1 stay essentially flat across distractor count within each dataset.

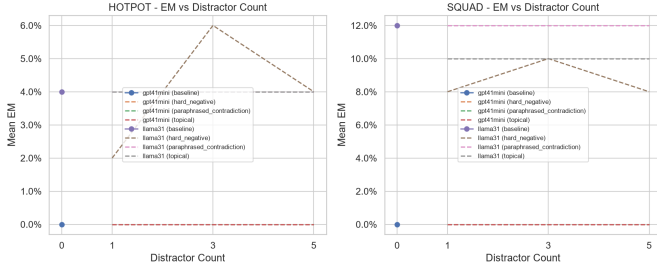


Fig. 2. Mean EM vs. distractor count, by dataset and distractor type.

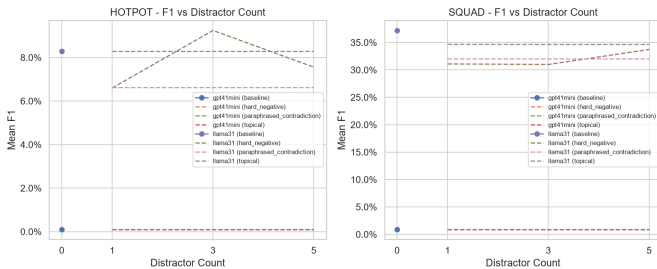


Fig. 3. Mean F1 vs. distractor count, by dataset and distractor type.

D. F1 Drop by Distractor Type

Fig. 4 shows mean F1 drop for Llama-3.1-8B Instant, averaged across both datasets, as a function of distractor type and count. Hard-negative distractors cause the largest mean F1 drop (0.039 at $k=1$, falling to 0.026 at $k=3$ and 0.021 at $k=5$). Paraphrased-contradiction distractors cause a constant 0.034 mean F1 drop regardless of count. Topical distractors cause the smallest,

count-invariant mean F1 drop of 0.013. This ordering – hard-negative > paraphrased-contradiction > topical – suggests lexical/semantic confusability at the retrieval boundary is the most disruptive perturbation type for this model, while purely topical noise is comparatively well tolerated, in line with the findings of Cuconasu et al. [4].

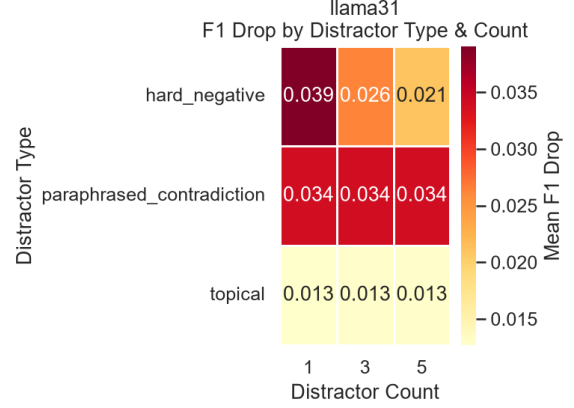


Fig. 4. Mean F1 drop by distractor type and count, Llama-3.1-8B Instant, averaged across HotpotQA and SQuAD.

E. Answer Flip Rate

For Llama-3.1-8B Instant, answer flip rates range from 0.00 to 0.06 across conditions (answer_flip_rate.csv), with hard-negative distractors on SQuAD producing the highest observed flip rate (0.06 at $k \in \{1, 3, 5\}$) and hard-negative distractors on HotpotQA the most count-sensitive pattern (0.02 at $k=1$, 0.00 at $k=3$, 0.02 at $k=5$). Paraphrased-contradiction and topical distractors produce lower flip rates on SQuAD (0.02, constant across counts) and zero flip rate on HotpotQA. GPT-4.1 mini shows a flip rate of 0.00 across all conditions, again attributable to the uniform API failure rather than genuine answer stability.

F. Retrieval Quality and Latency

Recall@5 and Recall@10 were invariant to distractor count and type within each (model, dataset) pair (overall_metrics.csv): 0.30/0.32 on HotpotQA and 1.00/1.00 on SQuAD for both models, showing that distractor injection perturbs the generator’s input context without touching the underlying retrieval-quality metrics, as intended by the framework design. Recorded per-condition latency was highly variable (from near-zero to several hundred milliseconds) and did not show a consistent monotonic relationship with distractor count in the

logged data; we do not draw a latency-scaling conclusion from these measurements and flag a controlled latency study as future work.

V. DISCUSSION

The results show the framework doing what it was built to do: FFS and the underlying F1-drop measurements differentiate distractor types in the direction predicted by prior noise-taxonomy work [4], with hard-negative (retrieval-boundary) distractors producing the largest hit and paraphrased-contradiction distractors producing a persistent, count-invariant faithfulness cost, while purely topical distractors are comparatively well tolerated by Llama-3.1-8B Instant. The non-monotonic hard-negative trajectory – degradation peaking at $k=3$ rather than $k=5$ on both datasets – echoes the RGB finding [5] that noise robustness is not a single monotone function of noise quantity, and is part of why fragility is worth measuring as a curve across distractor counts rather than at one fixed operating point, the way RAGChecker’s static Noise Sensitivity metric [9] does.

The GPT-4.1 mini condition is a genuine limitation of this dataset, not a substantive finding: because every call to the OpenRouter endpoint failed with an HTTP 402 insufficient-credit error, the recorded EM/F1/FFS of (near-)zero for this model reflects total generation failure, not observed fragility, and its mean FFS of 0.000 should not be read as evidence of superior robustness. Re-running the pipeline with a funded API key is necessary before any GPT-4.1 mini vs. Llama-3.1-8B Instant comparison can be made. This also shows a useful property of the framework: because FFS is computed straight from the same experiment logs regardless of which generator is used, a data-quality failure like this one is visible directly in the reported metrics instead of getting absorbed into an opaque aggregate score.

Recall@5/Recall@10 invariance across distractor conditions confirms the injection pipeline perturbs only the generator-facing context, keeping faithfulness fragility separate from retrieval-quality confounds, in line with the RGB goal of separating noise robustness from retrieval robustness proper [5].

A. Limitations

The evaluation reported here covers two generator models, one of which produced no valid generations due to an infrastructure failure external to the framework; 100 questions per dataset; and two QA benchmarks. FFS weights α, β in Eq. 2 were fixed for this run rather than swept, and confidence intervals in the style of ARES [8]

were not computed for the reported means. Latency measurements were not controlled for caching or network variance and should not be used for throughput claims. Human validation of FFS against annotator judgments, in the style of Adlakha et al. [10] and RAGTruth [11], was not performed in this run and is left as future work.

VI. CONCLUSION

We presented Distractor Injection, a reproducible, adversary-agnostic distractor-injection framework spanning three distractor types (topical, hard-negative, paraphrased-contradiction) at four controllable distractor counts, and the Faithfulness Fragility Score (FFS), a metric that reports the rate of faithfulness collapse under distractor pressure rather than a single value at one fixed noise condition. On SQuAD v2 and HotpotQA, Llama-3.1-8B Instant obtains a mean FFS of 0.124, with hard-negative distractors producing the largest mean F1 drop (0.039) and topical distractors the smallest and most count-invariant (0.013). The GPT-4.1 mini condition failed uniformly at the API layer and is reported transparently as a data-quality limitation rather than a robustness finding. The framework itself – its perturbation pipeline, distractor taxonomy, and FFS formulation – is meant as a reusable, domain-agnostic robustness-probing methodology that future work can apply to additional generator models, retrievers, and QA domains.

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